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Saveetha Institute of Medical And Technical Sciences
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A CAPSTONE PROJECT REPORT

Occupancy-Aware Smart Library Seat Reservation

with Predictive Demand Analytics

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

CSA0925-PROGRAMMIN IN JAVA

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

AIML

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Occupancy-Aware Smart Library Seat Reservation with Predictive Demand Analytics” has been carried out by L. Jayakannan(192525376) under the supervision of Guide Name and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I L. Jeyakannan of the AI&ML Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Occupancy-Aware Smart Library Seat Reservation with Predictive Demand Analytics” is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original and has been carried out in accordance with engineering ethics and academic integrity. All sources, references, software, datasets and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis and findings presented in this report have not been fabricated, falsified or manipulated.

Any external tools, computational resources or AI-assisted tools used during the project have been used responsibly and verified by me. I confirm that the applicable academic, institutional and professional requirements have been followed throughout the planning, design, implementation, testing and documentation of the project.

Place: Chennai

Date: September 2026

Signature of the student

L. Jeyakannan(192525376)

INDIVIDUAL CONTRIBUTION STATEMENT

Students: L. Jeyakannan(192525376)

Table: Individual Contribution

Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Database design, occupancy analysis, recommendation support, testing and documentation	Contributed to database structure, occupancy/demand analysis, recommendation logic and supporting implementation.	Contributed to test execution, result verification, edge-case checking and analysis of module outputs.	Prepared literature review, implementation/results sections and final documentation support.	50%

ABSTRACT

University libraries often experience uneven seat demand, especially during peak study periods. Students may spend time searching for vacant seats, while some seats remain underused because availability and reservations are not coordinated. Manual seat allocation also provides limited information for predicting future demand. These issues create an engineering problem involving resource availability, time-based demand variation, reservation conflicts and efficient use of library space.

The proposed Occupancy-Aware Smart Library Seat Reservation with Predictive Demand Analytics is designed as a desktop-based smart reservation solution using Java AWT for the user interface, Java application logic for processing, JDBC for database connectivity and MySQL for persistent storage. The system allows users to view seat availability, reserve an appropriate seat, manage reservation periods and receive demand-oriented recommendations. Historical occupancy and reservation information are organized so that demand patterns can be analysed for different time periods and used to support future planning.

The development approach follows an iterative engineering process involving requirement identification, database and interface design, module implementation, validation and documentation. Major modules include authentication, seat management, reservation, occupancy tracking, predictive demand analysis, recommendation, user feedback and reporting. The design intentionally avoids dependence on physical IoT sensors so that the core system can be developed, tested and demonstrated using software-based records and historical data.

The expected engineering outcome is a practical library seat management system that improves visibility of seat availability, reduces reservation conflicts and provides decision support through historical demand analysis. The system also provides a foundation for later integration with real-time sensors, mobile interfaces and more advanced machine-learning models. Actual performance values and acceptance results should be inserted from the final project testing evidence before submission.

Keywords: Smart Library, Seat Reservation, Occupancy Analysis, Predictive Demand Analytics, Java AWT, MySQL

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AWT	Abstract Window Toolkit	—
JDBC	Java Database Connectivity	—
DBMS	Database Management System	—
UI	User Interface	—
ML	Machine Learning	—
ER	Entity Relationship	—
UML	Unified Modeling Language	—
SQL	Structured Query Language	—
CRUD	Create, Read, Update, Delete	—
SDG	Sustainable Development Goal	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1. Background

Academic libraries provide essential study space for students, researchers and faculty. During normal periods, students can usually locate seats without difficulty, but demand becomes uneven during examinations, assignment deadlines and other peak periods. A conventional approach depends on students physically checking available seats or relying on manual reservation procedures. This creates unnecessary movement, uncertain availability and inefficient use of limited study space.

An occupancy-aware digital reservation system can centralize seat information and provide a structured way to manage availability. When reservation history and occupancy records are retained, the same information can also be analysed to identify recurring demand patterns. This creates an opportunity to move from simple seat booking towards demand-aware planning and recommendation.

1.2 Need for the Project

The project is needed because library seating is a limited shared resource. Students are affected when they cannot quickly determine which seats are available or when reservations conflict with actual availability. Library administrators also need better information about busy periods, underused areas and recurring demand. From an engineering perspective, the problem combines user interaction, resource-state management, time-based reservation constraints, persistent data storage and predictive analysis.

1.3 Problem Statement

The engineering problem is to design and implement a software-based smart library seat reservation system that maintains seat availability, validates reservations, records occupancy-related information and analyses historical demand to support better seat allocation and library planning. The system must provide a usable interface, consistent database state, conflict-aware reservation processing and an extensible prediction mechanism without requiring physical IoT infrastructure for the core implementation.

1.4 Project Aim

To develop an occupancy-aware smart library seat reservation system that combines digital seat booking with predictive demand analytics to improve seat visibility, reservation management and utilization planning.

1.5 Project Objectives

- To design a user-friendly Java AWT interface for library seat management and reservation.
- To develop a database-backed mechanism for maintaining users, seats, reservations and occupancy records.
- To implement reservation validation that checks seat availability and prevents conflicting bookings.
- To analyse historical occupancy and reservation data to identify time-based demand patterns.
- To implement a demand-oriented recommendation function that assists users in selecting suitable seats or periods.
- To test the major system functions and validate their behaviour against defined requirements.

1.6 Scope

The project includes user authentication, seat and zone management, seat availability display, reservation creation and validation, occupancy record handling, historical demand analysis, recommendation, reporting and database persistence. The core implementation is software based and can be demonstrated without physical occupancy sensors.

The project excludes hardware sensor deployment, biometric identification, payment processing, mobile application deployment and large-scale production infrastructure. These may be considered as future extensions. The prediction component is also treated as a decision-support feature; final prediction accuracy must be established from the project dataset and testing evidence rather than assumed.

1.7 Expected Engineering Outcome

The expected outcome is a working desktop application that provides a structured workflow from authentication to seat discovery, reservation and demand analysis. The system should maintain consistent seat and reservation states in the database and produce understandable demand summaries that can support both student decisions and library management planning.

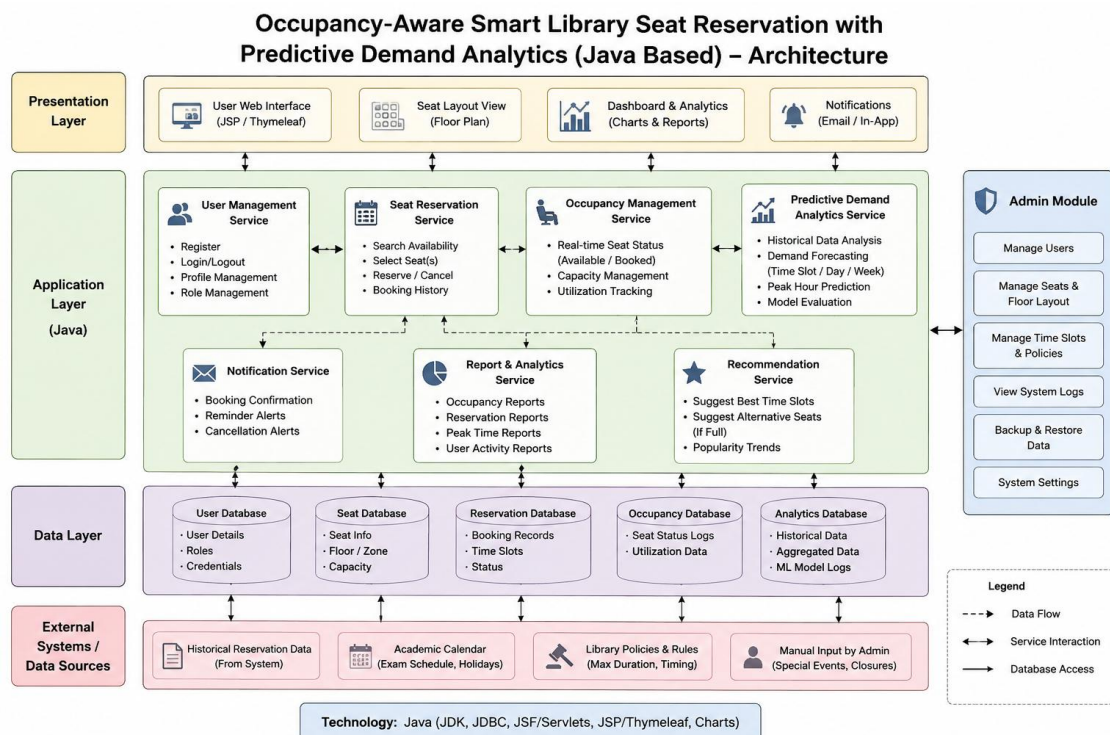


Figure 1: System Architecture

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review considered research on smart library seat management, occupancy monitoring, digital reservation systems and data-driven prediction of seating behaviour. Sources were selected from technical and academic publications that describe either software-based reservation approaches, occupancy monitoring systems or machine-learning analysis of library seating.

2.2 Review of Existing Work

Research on library seat management shows that seat allocation can be improved by making occupancy information visible and allowing users to reserve study spaces. Zhang, Bai and Fan described an intelligent library seat selection approach using sensors and QR-code-based information management to simplify seat selection and improve real-time awareness. Maepa and Moeti similarly investigated an IoT-based seat occupancy and reservation system using RFID and force-sensitive resistor technologies, demonstrating the feasibility of real-time seat monitoring.

Other smart-library research extends beyond seat reservation. Jayawardena et al. proposed an AI-based smart library management system that combines several library functions with intelligent recommendations and prediction-oriented components. A systematic review by Maepa and Moeti identified recurring challenges in smart library seat occupancy and reservation, including accurate occupancy information, reservation methods and estimation of reservation time.

More recent work has examined seat-level behaviour using machine-learning methods. A 2025 study based on long-term occupancy observations from an academic library analysed spatial, environmental and temporal attributes to understand seating preferences. The work demonstrates the value of historical occupancy data for predictive analysis, while also showing that human seating behaviour is complex and cannot be represented by a single simple factor.

2.3 Comparative Analysis

Table 1: Comparative Analysis of Existing Approaches

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual seat observation / first-come use	Simple and requires little technology	Poor visibility, time wastage and no historical analytics	Problem baseline
Sensor-based smart seating	Real-time occupancy awareness	Requires hardware deployment and maintenance	Useful future extension
Online seat reservation	Digital booking and improved visibility	May not provide demand prediction	Directly relevant
AI / ML smart library systems	Supports recommendations and prediction	Can be complex and data dependent	Supports predictive module
Proposed software-based system	Reservation + occupancy history + demand analytics without mandatory IoT	Prediction quality depends on available historical data	Target contribution

2.4 Research / Engineering Gap

Existing solutions frequently focus on either real-time occupancy sensing or reservation management, while predictive analysis may be treated as a separate capability. Sensor-based systems also introduce hardware cost and deployment dependencies. The identified gap is therefore a software-centred system that integrates reservation validation, occupancy history and demand-oriented analytics in one workflow while remaining implementable and testable without mandatory IoT hardware.

2.5 Proposed Contribution

The proposed project integrates seat reservation with historical occupancy and demand analysis. Instead of treating booking as an isolated transaction, the system maintains records that can be used to identify peak periods and support recommendations. The architecture also separates the interface, application logic and database so that a more

advanced machine-learning model or real-time sensor source can be incorporated later without redesigning the complete system.



Figure 2: System Flow

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Table 2: Stakeholder Requirements

Stakeholder	Requirement
Student / Library User	View available seats, select a suitable seat and make or cancel a reservation.
Library Administrator	Manage seats, zones, users and reservation records.
Library Management	View occupancy and demand summaries for planning.
Project Administrator / Developer	Maintain database, configuration and system reliability.

Functional & Non-Functional Requirements

Table 3: Functional and Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	User authentication	Valid users can access the dashboard; invalid credentials are rejected.
Functional	Seat availability	Current stored seat status is displayed correctly.
Functional	Reservation	Available seats can be reserved for a valid period.
Functional	Conflict validation	Overlapping reservation for the same seat is rejected.
Functional	Demand analytics	Historical records can be grouped by relevant time periods.
Functional	Recommendation	The system can produce a

		demand-aware seat/time recommendation.
Non-Functional	Usability	Common operations should be understandable through the AWT interface.
Non-Functional	Reliability	Database operations should preserve consistent reservation and seat states.
Non-Functional	Security	User-supplied database values should be handled using parameterized queries.
Non-Functional	Maintainability	UI, application logic and database access should remain separable.

3.2 Constraints & Applicable Standards

The system is constrained by the selected Java AWT desktop environment, MySQL database availability, JDBC connectivity, limited project development time and the absence of mandatory physical occupancy sensors. The design follows general software engineering and secure database practices. IEEE-style citation and documentation conventions are used for the report. Security implementation focuses on authentication, input validation and parameterized database operations rather than unsupported claims of complete security.

Table 4: Applicable Standards and Practices

Standard / Practice	Requirement	How Applied in This Project
IEEE referencing convention	Consistent citation of technical literature	Used for literature and reference formatting.
SQL/JDBC secure query practice	Avoid unsafe string concatenation for user input	PreparedStatement is used for parameterized operations.
UML modelling convention	Represent system structure and interactions clearly	Use case, class, activity and architecture diagrams are planned for the design chapter.

3.3 Design Specifications

Table 5: Design Specifications

Parameter	Target / Requirement	Unit	Priority
GUI technology	Java AWT only	Technology	High
Database	MySQL	Technology	High
Connectivity	JDBC / MySQL Connector	Technology	High
Reservation conflict	No overlapping confirmed booking for same seat	Condition	High
Occupancy data	Store time-associated occupancy information	Record	High
Demand analysis	Group historical demand by time/period	Analysis	High
Authentication	Validate user credentials before dashboard access	Condition	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Sensor-based smart library. This approach uses physical occupancy sensors and a network layer to obtain real-time seat status. It offers strong real-time visibility but increases hardware, deployment and maintenance requirements.

Alternative 2: Software-only reservation and analytics. This approach stores seat, reservation and occupancy information in a database and performs analysis through the application. It is easier to develop and demonstrate within the project constraints.

Alternative 3: Web/mobile-first system. This approach provides broader device access but requires additional web/mobile technologies and deployment infrastructure beyond the core Java AWT scope.

Table 6: Alternative Solution Evaluation

Criterion	Weight	Alternative 1: Sensor	Alternative 2: Software	Alternative 3: Web/Mobile
Performance /	High	High	Medium	High

real-time awareness				
Cost / project feasibility	High	Low	High	Medium
Safety / deployment simplicity	Medium	Medium	High	High
Sustainability / maintenance	Medium	Medium	High	Medium
Reliability of stored records	High	High	High	High

3.5 Selected Design & Detailed Design

The software-based architecture is selected because it satisfies the project objective while remaining feasible without physical sensor infrastructure. The system is organized into presentation, application and persistence layers. AWT screens collect user actions; Java business logic validates reservations and computes demand-related information; JDBC transfers operations to MySQL; and stored records provide the basis for occupancy and demand analysis.

System interface flow: User → AWT Interface → Validation / Business Logic → JDBC → MySQL → Result → AWT Output.

The major subsystem interfaces are: authentication with user records; seat management with seat/zone records; reservation with seat and time constraints; occupancy analytics with historical records; recommendation with demand analysis; and reporting with aggregated database queries.

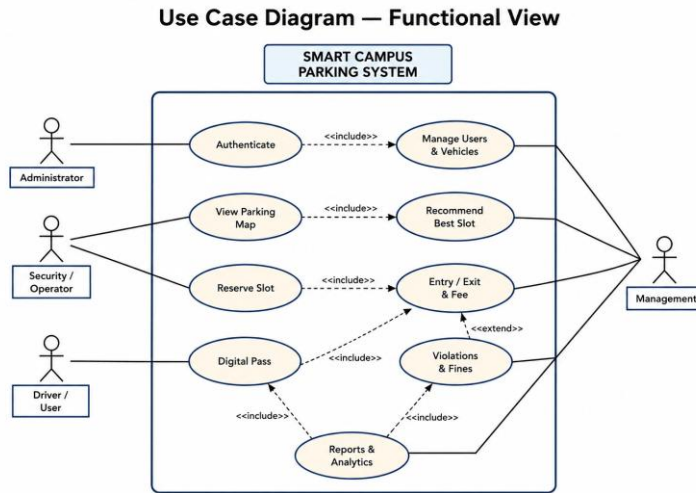


Figure 3:Case Diagram

3.6 Trade-off Analysis

Table 7: Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Java AWT interface	Swing / web UI	Matches project GUI requirement and native Java environment	Less modern than web UI	Selected because the implementation is intended for Java AWT.
MySQL persistence	File-based storage	Structured queries and reliable relational records	Requires DB setup	Selected for relational reservation and occupancy data.
Software occupancy records	Mandatory IoT sensors	Lower deployment cost and simpler testing	Not truly real-time without an external feed	Selected for the project scope; sensor integration is future work.
Historical demand baseline	Complex ML-only solution	Easy to validate and explain	May not capture complex behaviour	Selected as an extensible baseline; advanced ML can be added later.

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 8: Sustainability, Ethics, Safety and Security Considerations

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Avoid unnecessary hardware for core prototype	Project scope and available development resources	Software-first design reduces deployment dependencies.
Ethics	Avoid exposing unnecessary student information	Authentication and database access requirements	Only required user data should be stored and displayed.
Health & Safety	Reduce unnecessary movement while searching for seats	Problem context and reservation workflow	Digital availability reduces repeated physical searching.
Security	Protect authentication and database operations	Parameterized JDBC queries and validation	Use validation and restricted access to database functions.

Table 9: Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect reservation conflict logic	Medium	High	High	Validate time overlap before insertion and test conflict cases.	Low/Medium
Database connection failure	Medium	High	High	Handle exceptions, validate connection and display clear errors.	Medium
Insufficient historical data for prediction	Medium	Medium	Medium	Use a baseline analysis and clearly state data limitations.	Medium
Unauthorized account access	Medium	High	High	Authentication, validation and controlled database access.	Medium

Incorrect or stale occupancy records	Medium	Medium	Medium	Timestamp records and provide administrative correction/validation.	Medium
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ER DIAGRAM – SMART LIBRARY SEAT RESERVATION SYSTEM

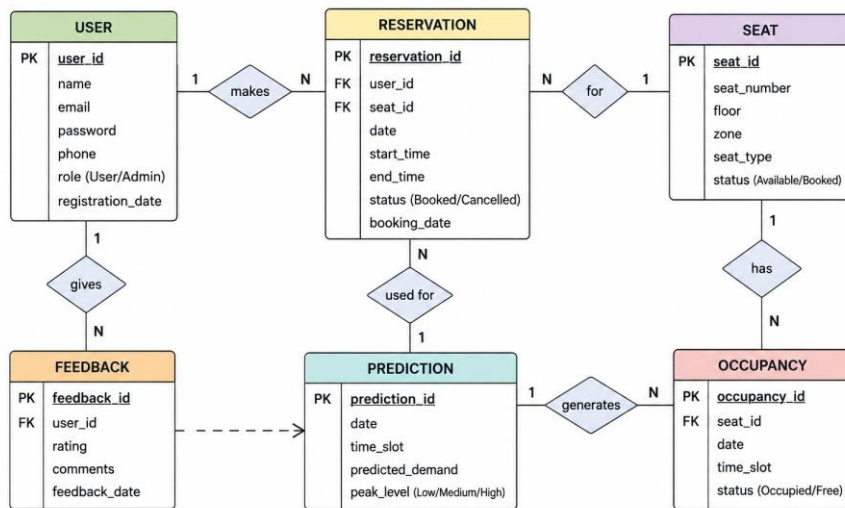


Figure 4:ER DIAGRAM

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

An iterative development approach is suitable because the system contains several connected modules whose behaviour can be validated incrementally. The development sequence is requirements analysis, database design, AWT interface development, reservation and occupancy logic, demand analytics, integration, testing and documentation. Each major module can be tested independently before full-system integration.

Table 10: Development Resources

Resource	Purpose
Java JDK	Compilation and execution of the application
Eclipse IDE	Java project development and debugging
Java AWT	Desktop graphical user interface
MySQL Server	Persistent data storage
MySQL Workbench	Database creation, inspection and query testing
MySQL Connector/J	JDBC connectivity between Java and MySQL

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The project is software-based. No fabrication is required for the core implementation. The application is implemented as a Java AWT desktop system connected to MySQL through JDBC. The main implementation areas are authentication, seat and zone management, reservation, occupancy records, demand analytics, recommendation and reporting.

Table 11: Software and Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Java	Application programming	Implementation

Java AWT	GUI construction and event handling	Front-end implementation
JDBC	Database connectivity	Back-end integration
MySQL	Relational database	Database implementation
MySQL Workbench	Database design and verification	Database testing
Eclipse IDE	Coding and debugging	Development
Git / GitHub	Version control and repository management	Documentation / submission

4.3 Test Plan & Verification

Table 12: Test Plan and Verification

Requirement	Test Method	Acceptance Criterion	Specification Required Value	Achieved Value	Status
Authentication	Enter valid and invalid credentials	Valid user reaches dashboard; invalid user is rejected	As defined	To be entered from actual test	To be entered
Seat availability	Open seat management/reservation view	Correct stored status is shown	As defined	To be entered from actual test	To be entered
Reservation	Book an available seat	Reservation record is created	As defined	To be entered from actual test	To be entered
Conflict validation	Book same seat with overlapping period	Conflicting booking is rejected	As defined	To be entered from actual test	To be entered
Demand analytics	Run demand report	Grouped demand summary is displayed	As defined	To be entered from actual test	To be entered
Recommendation	Request suitable	Recommendation	As defined	To be	To be

	seat/time	is generated from available data		entered from actual test	entered
Database integration	Compare GUI action with MySQL record	Application and DB state agree	As defined	To be entered from actual test	To be entered

4.4 Results

The final results section must contain evidence from the implemented system. The expected functional evidence includes successful login, seat availability display, reservation confirmation, conflict rejection, occupancy/demand analysis and recommendation output. Numerical performance values, prediction accuracy and other measurements must be entered only after executing the corresponding tests on the final implementation.

Recommended evidence to insert: (1) Eclipse project structure, (2) login screen, (3) main dashboard, (4) seat map / availability screen, (5) reservation confirmation, (6) demand analytics output, (7) recommendation output, and (8) MySQL Workbench records.

4.5 Data Analysis & Engineering Judgment

The central engineering judgement is that historical seat-use data is useful only when it is recorded with sufficient context, such as date, time period, seat/zone and occupancy or reservation status. Aggregating these records can reveal recurring peak periods and relatively low-demand periods. However, predictions are dependent on data quality, sample size and changes in student behaviour. Therefore, the project should report actual model or baseline performance only after testing and should distinguish between observed historical demand and predicted demand.

If the observed results differ from the expected behaviour, possible causes include incomplete records, incorrect reservation timestamps, insufficient historical observations or unexpected user behaviour. These factors should be discussed with reference to the actual test evidence.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 13: Objective Achievement

Objective	Evidence	Achievement
Design a Java AWT seat reservation interface	Implemented AWT screens and event handling	To be verified from final screenshot
Maintain seats, users and reservations in MySQL	Database schema and JDBC operations	To be verified from DB evidence
Prevent reservation conflicts	Conflict validation test case	To be verified from test result
Analyse historical occupancy/demand	Demand analytics module and report	To be verified from output
Provide demand-oriented recommendation	Recommendation module/output	To be verified from output
Test major functions	Test plan and executed results	To be verified from completed test table

4.7 Strengths & Limitations

Strengths:

- Integrates reservation and demand analysis in one workflow.
- Uses a relational database for persistent and traceable records.
- Can be developed and demonstrated without mandatory IoT hardware.
- Provides a clear path for future machine-learning and real-time occupancy extensions.

Limitations:

- Core occupancy status depends on software records unless an external real-time source is added.
- Prediction quality depends on the quantity and quality of historical data.
- Java AWT provides desktop access rather than a browser/mobile interface.
- Final performance and prediction metrics require evidence from actual project testing.

4.8 Project Planning, Roles & Budget

Table 14: Project Milestones

Milestone	Planned Activity	Expected Deliverable
1	Requirement analysis	Problem definition and requirements
2	System and database design	Architecture, UML and ER design
3	AWT interface development	Working user interface
4	Reservation and database integration	Core transaction workflow
5	Occupancy and demand analytics	Analytics module
6	Testing and debugging	Verified test cases
7	Documentation and presentation	Final report and demonstration

Table 15: Team / Individual Contribution

Student	Assigned Responsibility	Actual Contribution
E. Lakshmi Narayanan	System design, Java AWT, database integration, reservation and analytics documentation	To be confirmed against final GitHub/project evidence

Table 16: Project Cost

Item	Estimated Cost	Actual Cost
Existing development computer	—	—
Java / Eclipse development tools	No additional project cost expected	—
MySQL / Workbench	No additional project cost expected	—
IoT hardware	Not required for core implementation	—

Plate Number

Tn12322

Owner User ID

3

Vehicle Type

Car

ADD VEHICLE

DELETE PLATE

REFRESH

VEHICLE MANAGEMENT

CRUD

ID | USER ID | PLATE | TYPE

1 | 1 | TN01AA1234 | Car

2 | 2 | TN01BB5678 | Bike

3 | 2 | TN01CC9999 | Car

18 | 3 | TN463443 | Car

Figure 5:Interaction Interface

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Occupancy-Aware Smart Library Seat Reservation with Predictive Demand Analytics project addresses the practical problem of managing limited library study seats. The proposed solution combines digital reservation, seat availability management, historical occupancy information and demand-oriented analysis within a single software architecture. Java AWT provides the desktop interface, Java application logic handles validation and processing, JDBC provides database connectivity and MySQL maintains persistent records.

The engineering value of the project lies in connecting reservation management with analytical use of historical data rather than treating seat booking as an isolated transaction. The design also considers realistic constraints such as limited development resources, the need for reliable database state and the absence of mandatory IoT infrastructure. Final claims about performance, prediction accuracy and user acceptance should be based on the completed test evidence.

5.2 Future Work

- Integrate real-time occupancy sensors or other trusted occupancy sources.
- Develop a mobile or web interface for wider user access.
- Use advanced machine-learning models for time-based demand forecasting after sufficient data is collected.
- Add QR-based check-in and check-out to reduce unused reservations.
- Introduce automated notifications for reservation start, expiry and cancellation.
- Add richer dashboards for library management and long-term space planning.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Java AWT event-driven interface development
- JDBC and relational database integration
- Reservation conflict and state-management logic
- Occupancy data organization and demand analysis
- Software testing, documentation and system-level design

Engineering & Professional Skills Developed:

- Requirement analysis and technical documentation
- Database design and SQL problem solving
- Debugging and validation
- System decomposition and module integration
- Technical communication and project planning

Individual Reflection – E. Lakshmi Narayanan:

The most challenging part of the project was coordinating seat availability, reservation time ranges and database state so that a reservation could not create an inconsistent seat condition. I addressed this by separating validation from database operations and by considering conflict checks before confirming a reservation. A key engineering decision was to keep the core implementation software based rather than depending on physical sensors, which made the system easier to develop and test within the project scope. I also gained practical knowledge of connecting a Java AWT application to MySQL through JDBC and learned that predictive analysis must be supported by adequate historical data rather than assumed accuracy. If the project were repeated, I would collect a larger and more structured occupancy dataset and evaluate multiple prediction models before selecting the final approach.

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Note: Verify all bibliographic details against the final source copies before submission.

APPENDICES

Appendix A – Detailed Calculations

Insert any final demand-analysis calculations, metric calculations or other essential derivations used by the implemented system.

Appendix B – Engineering Drawings / Schematics

Insert the final system architecture, workflow, use case, class, activity and ER diagrams here if they are not placed in Chapter 3.

Appendix C – Source Code / Code Repository Information

Insert the final GitHub repository URL and selected essential code excerpts. Complete source code should be maintained in the approved repository.

Appendix D – Datasheets

Not applicable for the software-only core implementation unless external hardware is added.

Appendix E – Experimental / Raw Data

Insert the occupancy/reservation dataset or a representative extract, together with data dictionary and preprocessing notes.

Appendix F – Ethics / Institutional Approval

Insert applicable approval or state not applicable according to departmental guidance.

Appendix G – Project Meeting / Progress Records

Insert meeting records, milestone evidence or progress screenshots where required.

Appendix H – User / Industry Feedback

Insert actual feedback forms or summary evidence if collected.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Insert evidence if applicable; otherwise mark as not applicable.

Appendix J – Individual Contribution Evidence

Insert GitHub contribution evidence, assigned tasks, screenshots or other verifiable contribution records as required by the department.